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When the loop starts rotating due to this torque, the magnetic field $\vec{B}$ is no longer in the plane of the loop. So the above equation is the special case.

When the loop starts rotating about z axis due to this torque, the magnetic field $\vec{B}$ is no longer in the plane of the loop. So the above equation is the special case.

(ii) when unit vector $\hat{n}$ is at an angle θ with the field

In the general case, the unit normal vector $\hat{n}$ and magnetic field $\vec{B}$ is with an angle θ as shown in Figure 3.63.

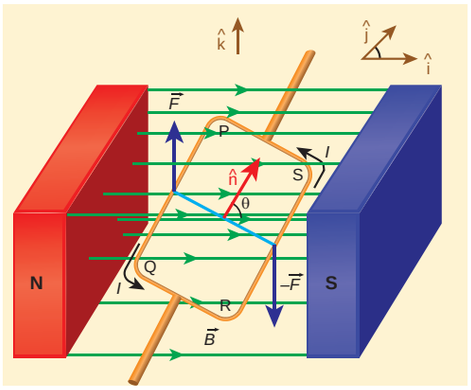


Figure 3.63 Unit vector makes an angle **θ** with the field

(a) The force on section PQ

$$\vec{l} = -a\,\hat{j} \;and\; \vec{B} = B\,\hat{i}$$

$$\vec{F}_{PQ} = I\vec{l} \times \vec{B} = -IaB\left(\hat{j} \times \hat{i}\right) = IaB\,\hat{k}$$

Since, the unit vector normal to the plane $\hat{n}$ is along the direction of $\hat{k}$.

(b) The force on section QR

$$\vec{l} = b\cos\left(\frac{\pi}{2} - \theta\right)\check{i} - \sin\left(\frac{\pi}{2} - \theta\right)\check{k} \;and\; \vec{B} = B\,\check{i}$$

$$\vec{F}_{QR} = I\vec{l} \times \vec{B} = -IbB\sin\left(\frac{\pi}{2} - \theta\right)\hat{j}$$

$$\vec{F}_{QR} = -IbB\cos\theta\,\hat{j}$$

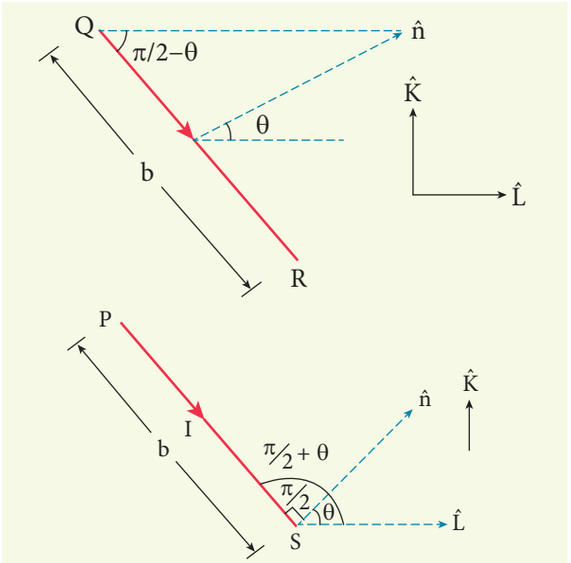


Figure 3.64 Horizontal and vertical component of the sections - (a) QR (b) SP

(c) The force on section RS

$$\vec{l} = a\,\check{j} and\, \vec{B} = B\hat{i}$$

$$\vec{F}_{RS} = I\vec{l} \times \vec{B} = IaB\left(\hat{j} \times \hat{i}\right) = -\mathrm{IaB}\,\hat{\mathrm{k}}$$

Since, the unit vector normal to the plane is along the direction of $-\hat{k}$.

(d) The force on section SP

$$\vec{l} = b\cos\left(\frac{\pi}{2} + \theta\right)\hat{i} + \sin\left(\frac{\pi}{2} + \theta\right)\hat{k} \;\; and \;\; \vec{B} = B\,\hat{i}$$

$$\vec{F}_{SP} = I\vec{l} \times \vec{B} = IbB\sin\left(\frac{\pi}{2} + \theta\right)\hat{j}$$

$$\vec{F}_{SP} = IbB\cos\theta\,\hat{j}$$

The net force on the rectangular loop is

$$\vec{F}_{net} = \vec{F}_{PQ} + \vec{F}_{QR} + \vec{F}_{RS} + \vec{F}_{SP}$$

$$\vec{F}_{net} = IaB\hat{k} - IbB\cos\theta\,\hat{j} - IaB\hat{k} + IbB\cos\theta\,\hat{j}$$

$$\Rightarrow \vec{F}_{net} = \vec{0}$$

Hence, the net force on the rectangular loop in this configuration is also zero. Notice that the force on section QR and SP are not

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